

Status_Summary

RK3588 / Orange Pi 5 Max — A/B-virt Emulator — Status Summary

Revision: 1 **Last modified:** 2026-06-11T12:10:00Z **Companion of:** [Status.md](#) (§11.4.56 two-audience parity).

Page 1 — For the operator (plain language)

We are building a way to test the RK3588 / Orange Pi 5 Max over-the-air (OTA) update flow **without needing the actual board on the desk yet** — by emulating the device on this Mac.

What works today:

- The emulated device **boots up and runs**. We built a small aarch64 Linux image and it boots all the way to a working login on this Mac (using Apple's built-in virtualization). We have the full boot recording as proof.
- The container-based test (control plane talking to a fake device) also works end-to-end.

What is still pending (NOT done yet — honestly):

- **The real “switch to the new system half and roll back if it fails” part is NOT proven yet.** That is the heart of the A/B update (pick slot A or B, verify it, fall back if the new one is bad). The pieces for it are being built into the image right now, but there is no proof it works yet, so we are not claiming it does.
- **The full real-Android A/B update test needs a Linux machine** (it cannot run on this Mac — the Mac lacks the required virtualization device). It is ready to run the moment your incoming Linux host is available; for now it is correctly **skipped**, not faked.
- Testing on the **real RK3588 / Orange Pi 5 Max board** is pending — no board on the bench yet.

Bottom line: boots on the Mac = proven. Real A/B slot switch + the Linux-host Android A/B test = pending / waiting on the right environment. We do not pretend anything works that has not been captured as evidence.

Page 2 — For software engineers

Fidelity ladder. T0 podman control-plane+emulator (SHIPPED, protocol-only) → T1 A/B-virt base image on QEMU virt + HVF (FOUNDATION GREEN: live-userspace boot; A/B-apply layers in progress) → T2 Cuttlefish real Android update_engine A/B + AVB/dm-verity + auto-rollback (SKIP on this host, Linux+KVM-gated) → T3 real RK3588 hardware (PENDING).

Proven (captured evidence):

- **PWU-AB-1 base image** — tests/emulator/ab_virt/out/images/Image (MD5 9f3670cd7dba7bdeeebe2c6d791e929a) + tests/emulator/ab_virt/out/images/rootfs.ext2 (MD5 a056760e88eea575977be13e38cfe430); builder tests/emulator/ab_virt/build_image.sh. (A U-Boot+RAUC rebuild is in flight and will replace these.)
- **PWU-AB-1 foundation boot** — docs/qa/20260611T061626Z-ab-virt-boot/console.log (196 lines): kernel on Apple CPU MIDR 0x610f (physical CPU ... [0x610f0000]), Buildroot buildroot login: root, post-login uname -a → Linux buildroot 6.1.44 ... aarch64, sentinel HELIX_USERSPACE_LIVE_OK (line 182), clean poweroff. §11.4.107 liveness battery (full transcript + post-login sentinel, not a single frame). Driver tests/emulator/ab_virt/boot_smoke.sh.
- **T0 round-trip** — tests/emulator/tier1_container_e2e.sh (register → update-check → telemetry asserted from captured podman logs under docs/qa/<run-id>/).

Pending / not proven (§11.4.6):

- **A/B slot switch + dm-verity + auto-rollback (E3–E5)** = PENDING_FORENSICS. No captured slot-switch or rollback evidence. U-Boot qemu_arm64 bootcount + RAUC dm-verity slot layers (commit 4278aa9) are being built into the image; not yet exercised.
- **T2 Cuttlefish (E7)** = SKIP. Driver tests/emulator/tier2_cuttlefish_ab.sh; host gate /dev/kvm absent on this Apple-Silicon macOS host (verified). Exit-3 honest topology SKIP per §11.4.3/§11.4.112. Script header marks the exact OTA-apply invocation UNCONFIRMED: (Virtual-A/B vs legacy A/B; update_device.py vs cvd) — runtime-detected, never guessed. Ready for the operator's incoming Linux + nested-KVM host.
- **T3 hardware (E8)** = PENDING_FORENSICS. No board.

Provenance: dd43738 (research + dev-host A/B-virt build infra) → d5374d0 (PWU-AB-1 foundation GREEN, live-userspace boot) → 4278aa9 (U-Boot + RAUC + GPT tooling, PWU-AB-1 full in progress). HEAD 4278aa9.

§-refs: §11.4.45 (Status doc), §11.4.56 (two-audience summary), §11.4.5 (captured-evidence table), §11.4.6 (no-guessing), §11.4.107 (liveness), §11.4.3 / §11.4.112 (topology-gated SKIP), §11.4.44 (revision header).